

Computational Intelligence: Topics for Internal Examination

1. Fuzzy sets - $P \cup Q, P \cap Q, P', Q', P - Q, P \oplus Q, P \cup P' \neq U$ and $P \cap P' \neq \emptyset$ etc.
2. Fuzzy logic - "computing with words".
3. genetic algorithms, *fitness-proportional selection* (roulette wheel selection with replacement) *Stochastic universal selection* – a variant of fitness-proportional selection.
4. Basic ant behavior that is modeled to create the artificial ant colony systems for solving optimization problems.
5. Mamdani fuzzy inference system.
6. General life cycle of a genetic algorithm – block diagram.
7. General life cycle of the ant colony optimization (ACO) algorithm.
8. The standard Particle Swarm Optimization (PSO) algorithm.